DEFENCE JOURNEY

NDA Geography Notes

Physical, Indian and world geography for the NDA GAT paper

NDA | GEOGRAPHY

Geography in NDA includes the Earth and its structure, weather and climate, India's physical features, resources and agriculture, and basic world geography. Learn the facts and use a map while revising.

1. The Earth and the Solar System

- The Earth is about 4.5 billion years old. It is the third planet from the Sun and the only one known to support life. Shape: **geoid** (slightly flattened at the poles).
- Planets in order: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune. Largest: Jupiter. Smallest: Mercury. Hottest: Venus. The Red Planet: Mars. Pluto is a dwarf planet.
- The Moon is Earth's only natural satellite. Light from the Sun takes about 8 minutes 20 seconds to reach the Earth.
- **Rotation** (24 hours) causes day and night. **Revolution** (365¼ days) with the tilt of the axis (23½°) causes the seasons. A leap year has 366 days.
- **Equinox** (21 March and 23 September): day and night are equal. **Summer solstice** (21 June): longest day in the northern hemisphere. **Winter solstice** (22 December): shortest day.
- **Solar eclipse:** the Moon comes between the Sun and the Earth (new moon day). **Lunar eclipse:** the Earth comes between the Sun and the Moon (full moon day).
- Latitudes run east-west (the Equator is 0°). Longitudes run north-south (the Prime Meridian, 0°, passes through Greenwich). The Earth turns 15° per hour, so 1° of longitude = 4 minutes.
- Indian Standard Time is based on 82°30′ E (near Mirzapur, Uttar Pradesh) and is 5 hours 30 minutes ahead of GMT. The International Date Line is near 180°.

2. Structure of the Earth and Landforms

- **Layers:** Crust (outermost, thinnest) → Mantle → Core (outer core liquid, inner core solid; iron and nickel).
- **Rocks:** Igneous (formed from cooled magma: granite, basalt), Sedimentary (layered, contain fossils: sandstone, limestone), Metamorphic (changed by heat and pressure: marble from limestone, slate from shale, gneiss from granite).
- **Plate tectonics:** the lithosphere is made of moving plates. Volcanoes and earthquakes mostly occur at plate boundaries. The Pacific **Ring of Fire** has most volcanoes.
- The point where an earthquake starts underground is the **focus**; the point above it on the surface is the **epicentre**. Seismographs record earthquakes, and the Richter scale measures magnitude.
- **Mountains:** Fold (Himalayas, Alps), Block (Vosges), Volcanic (Mount Fuji), Residual (Aravalli). Fold mountains are the youngest. The Himalayas are young, the Aravalli are among the oldest.
- **Plateaus:** Deccan plateau (volcanic, black soil), Tibetan plateau (the highest, called the Roof of the World). **Deltas** form at river mouths (Sundarbans delta, the largest in the world).

- **Rivers do:** erosion, transportation, deposition. Features: V-shaped valley, waterfall, meander, ox-bow lake, delta.

3. Atmosphere, Weather and Climate

Layer	Height	Key facts
Troposphere	0 to about 12 km	Weather happens here. Temperature falls with height.
Stratosphere	12 to 50 km	Contains the **ozone layer**. Jet aircraft fly here.
Mesosphere	50 to 80 km	Meteors burn here. Coldest layer.
Thermosphere (Ionosphere)	80 to 400 km	Reflects radio waves. Aurora occurs here.
Exosphere	above 400 km	Merges into space.

- Air composition: nitrogen 78%, oxygen 21%, argon about 0.93%, carbon dioxide about 0.04%.
- **Winds:** Trade winds (blow from the subtropical high to the equatorial low), Westerlies (30° to 60°), Polar easterlies. Local winds: Loo (hot, North India), Chinook (warm, Rockies), Mistral (cold, France).
- **Cyclones** have low pressure at the centre. Called hurricane (Atlantic), typhoon (Pacific, China), willy-willy (Australia), cyclone (Indian Ocean). The calm centre is the eye.
- Humidity is the amount of water vapour in the air. Rain types: convectional, orographic (relief), cyclonic.
- The **Greenhouse effect** keeps the Earth warm; excess greenhouse gases cause global warming.
- Instruments: thermometer (temperature), barometer (air pressure), hygrometer (humidity), rain gauge, anemometer (wind speed).

4. Oceans and World Geography

Item	Fact
Oceans (largest to smallest)	Pacific, Atlantic, Indian, Southern, Arctic. The deepest point is the Mariana Trench (Challenger Deep, about 11 km).
Continents (largest to smallest)	Asia, Africa, North America, South America, Antarctica, Europe, Australia.
Highest peak	Mount Everest (8,849 m, Nepal and China border).
Longest river	Nile (Africa) is generally regarded as the longest. The Amazon carries the most water.
Largest hot desert	Sahara (Africa). Largest cold desert: Antarctica. The Gobi is in Asia.
Largest island	Greenland. Largest lake (area): Caspian Sea. Deepest lake: Baikal (Russia).
Highest waterfall	Angel Falls (Venezuela).
Lowest point on land	Dead Sea (Jordan and Israel).
Largest country by area	Russia. Smallest: Vatican City. Most populous: India (the largest in the world).
Important straits	Strait of Malacca (Indian Ocean and Pacific), Strait of Hormuz (Persian Gulf), Bosporus, Strait of Gibraltar.
Canals	Suez Canal (Mediterranean Sea to Red Sea), Panama Canal (Atlantic to Pacific).

- **Ocean currents:** Warm: Gulf Stream, Kuroshio, North Atlantic Drift. Cold: Labrador, California, Humboldt (Peru). Meeting of warm and cold currents makes rich fishing grounds (Grand Banks).
- **Tides** are caused by the gravitational pull of the Moon and the Sun. Spring tides occur on new moon and full moon days.

5. Physical Geography of India

- India has **28 states and 8 Union Territories**. Area: about 3.28 million sq km, the 7th largest country in the world. Population (Census 2011): 121 crore, literacy 74%. Largest state by area: **Rajasthan**. Smallest: **Goa**. Most populous: **Uttar Pradesh**.
- The **Tropic of Cancer** passes through 8 states: Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram.
- Southernmost point: Indira Point (Great Nicobar). Northernmost: Indira Col. India has a 7,516 km coastline (mainland about 6,100 km).
- **Himalayas:** three parallel ranges: Himadri (Greater, highest peaks), Himachal (Lesser) and Shiwaliks (Outer). Kanchenjunga (8,586 m) is the highest peak in India. The Karakoram range has K2.
- **Northern Plains** are formed by alluvium of the Indus, Ganga and Brahmaputra. **Peninsular Plateau** is among the oldest landmasses. **Western Ghats** (Sahyadri) are higher than the Eastern Ghats. Anamudi (Kerala) is the highest peak in South India.
- **Thar Desert** is in Rajasthan. **Rann of Kutch** is in Gujarat. **Chilika Lake** (Odisha) is the largest coastal lagoon. Wular Lake (Kashmir) is the largest freshwater lake. Sambhar (Rajasthan) is the largest inland salt lake.
- **Islands:** Andaman and Nicobar (Bay of Bengal), Lakshadweep (Arabian Sea).

Major rivers

River	Origin	Flows into / Facts
Ganga	Gangotri (Uttarakhand), called Bhagirathi	Bay of Bengal. Longest river in India. Tributaries: Yamuna, Gandak, Kosi, Son.
Brahmaputra	Tibet (called Tsangpo)	Bay of Bengal. Majuli, the largest river island, is on it.
Indus	near Mansarovar, Tibet	Arabian Sea. Tributaries: Jhelum, Chenab, Ravi, Beas, Sutlej.
Godavari	Nashik (Maharashtra)	Bay of Bengal. Longest peninsular river (Dakshin Ganga).
Krishna / Kaveri	Western Ghats	Bay of Bengal. Kaveri has the Shivasamudram falls.
Narmada and Tapi	Amarkantak / Satpura	Flow westward into the Arabian Sea through rift valleys.
Mahanadi	Chhattisgarh	Bay of Bengal. Hirakud Dam (one of the longest dams).

6. Climate, Soil and Agriculture in India

- India has a **tropical monsoon climate**. Seasons: Winter (Dec to Feb), Summer (Mar to May), South-West Monsoon (June to Sept), Retreating Monsoon (Oct to Nov). Mawsynram and Cherrapunji (Meghalaya) have the highest rainfall. Rajasthan (Jaisalmer) has the lowest.
- **Soils:** Alluvial (most fertile and widespread, rice, wheat, sugarcane). Black or Regur (cotton, retains moisture, Deccan plateau). Red (iron oxide, acidic). Laterite (Kerala, tea and coffee). Desert soil (Rajasthan).
- **Crop seasons:** Kharif (June to Oct): rice, maize, cotton, jowar, bajra, soybean, groundnut. Rabi (Oct to March): wheat, barley, gram, mustard, peas. Zaid (summer): watermelon, cucumber, vegetables.
- **Cash crops:** cotton, jute, sugarcane, tea, coffee, rubber. India is the largest producer of **milk, pulses and jute**, and among the top producers of rice, wheat, sugarcane and tea.
- Tea: Assam, West Bengal (Darjeeling). Coffee: Karnataka (Coorg). Rubber and spices: Kerala. Wheat: Punjab, Haryana, UP. Cotton: Gujarat, Maharashtra.
- **Green Revolution** (1960s, M. S. Swaminathan, with Norman Borlaug's high-yielding wheat) increased wheat and rice output. White Revolution (milk, Verghese Kurien, Operation Flood). Blue Revolution (fish). Yellow Revolution (oilseeds).

7. Minerals, Industry and Transport

Resource	Main areas
Coal	Jharia and Dhanbad (Jharkhand), Raniganj (West Bengal), Singrauli, Korba (Chhattisgarh), Talcher (Odisha)
Iron ore	Odisha, Jharkhand, Chhattisgarh, Karnataka, Goa
Mica	Jharkhand (Koderma), Andhra Pradesh, Rajasthan
Bauxite	Odisha, Jharkhand, Gujarat
Petroleum	Mumbai High (offshore), Assam (Digboi, the oldest), Gujarat (Ankleshwar)
Copper	Khetri (Rajasthan), Singhbhum (Jharkhand), Balaghat (MP)
Steel plants	Bhilai, Rourkela, Bokaro, Durgapur, Jamshedpur (private, TISCO), Salem, Vishakhapatnam

- India's major ports: Mumbai (largest natural harbour), Jawaharlal Nehru Port (Nhava Sheva), Kandla, Chennai, Kolkata, Visakhapatnam, Paradip, Kochi, Mormugao.
- Largest railway network in Asia; Indian Railways is among the largest in the world. Golden Quadrilateral links Delhi, Mumbai, Chennai and Kolkata. The National Highway 44 (Srinagar to Kanyakumari) is the longest.
- Hydroelectric dams: Bhakra Nangal (Sutlej, Himachal Pradesh), Tehri (the highest in India, Bhagirathi), Hirakud (Mahanadi), Sardar Sarovar (Narmada), Nagarjuna Sagar (Krishna), Mettur (Kaveri).
- Cities: Bengaluru (IT hub), Mumbai (financial capital), Kolkata, Jamshedpur (Steel City), Ahmedabad (textiles, called Manchester of India), Kanpur (leather).

TIP: Practise with a blank outline map of India: mark rivers, mountain ranges, ports, plateaus and mineral belts. Maps stay in memory far longer than lists.